

SIH 26227 – System Workflow & Project Implementation

Technical Architecture, End-to-End Workflow, Current State & Next Development Stage

PROJECT TITLE: SIH Problem Statement 26227	DOMAIN: Remote Sensing & Geospatial Intelligence
SUB-TITLE: Semantic Retrieval & Multi-Temporal Change Analysis	TARGET MILESTONES: PPT: Sep 16, 2026 Hackathon: Sep 20, 2026
REPOSITORY STATE: Phase 1 STAC Verified (Zero Compile Errors)	DOCUMENT AUDIENCE: College Faculty, Evaluators & Technical Judges

FACULTY / JUDGE EXECUTIVE BRIEFING

This technical brief provides an honest, comprehensive evaluation of the **SIH 26227 codebase**. It details the exact communication bridges between the React frontend and FastAPI backend, documents the newly integrated **Copernicus Data Space STAC client** retrieving live Sentinel-2 Level-2A metadata, and contrasts the currently operational deterministic analysis engine against the planned pixel-level raster difference pipeline. No prior knowledge of the internal code repository is required to understand this workflow.

1. PROJECT OVERVIEW & PROBLEM DEFINITION

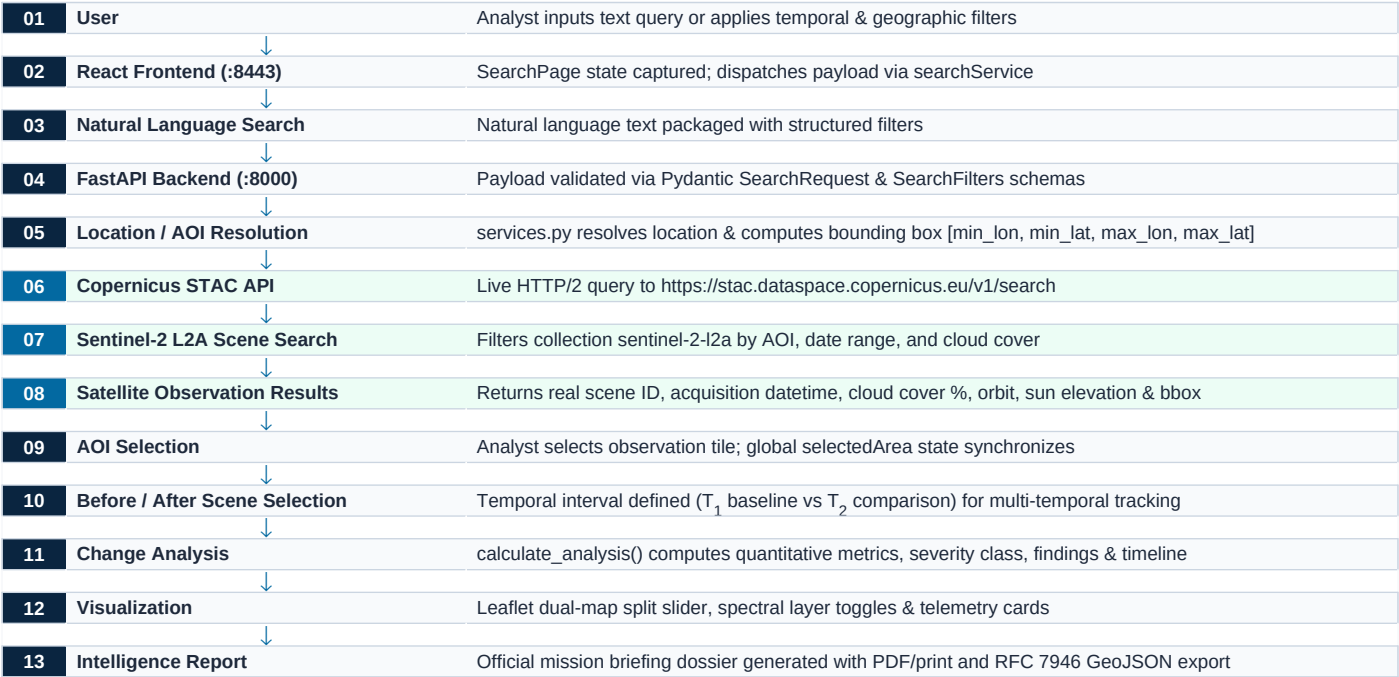
Smart India Hackathon Problem Statement **26227** addresses the national requirement for automated, reliable satellite intelligence: *'Semantic Retrieval and Multi-Temporal Change Analysis of Satellite Imagery'*. Earth observation data generated by constellations like Sentinel-2, Landsat-8, and Cartosat produce petabytes of imagery. Traditional GIS exploration requires specialized desktop software, manual tile identification, cloud masking, and offline raster differencing. Our platform bridges this gap through an end-to-end web system designed for defense, urban planning, and environmental surveillance.

End-to-End User Experience Lifecycle:

1. Query Submission	The user inputs a natural language prompt (e.g., 'Urban expansion in Hyderabad' or 'Forest loss near Nagaland') or selects structured filters (geographic state, date interval, change type, sensor).
2. Location & AOI Resolution	The FastAPI backend parses the query, matches coordinates against Indian geographic catalogs or curated regions, and computes an Area of Interest (AOI) bounding box.
3. Live Sentinel-2 STAC Search	The backend initiates an asynchronous HTTP/2 request to the Copernicus Data Space STAC API (collection: sentinel-2-l2a), querying surface reflectance scenes matching the AOI, acquisition dates, and cloud cover criteria.
4. Observation Display & Map View	The frontend visualizes retrieved scenes as interactive cards and Leaflet map markers, showing acquisition date, cloud percentage, sun elevation, MGRS tile, and orbit.
5. AOI Selection & Analysis Dispatch	The analyst selects a specific observation tile, triggering synchronization across the application state and opening the multi-temporal change detection engine.
6. Change Analysis & Quantitative Vectors	The backend processes the AOI parameters to generate change vectors (built-up expansion, water recession, canopy degradation), severity ratings, findings, and historical timelines.
7. Multi-Modal Intelligence Visualization	Results are visualized through interactive split-screen comparison sliders, spectral index toggle layers (NDVI, NDWI, false-color), and quantitative telemetry charts.
8. Mission Intelligence Reporting	The analyst exports findings as standardized GeoJSON polygons (RFC 7946) or printable formal intelligence dossiers for executive briefing.

2. MAIN SYSTEM WORKFLOW (END-TO-END DATA FLOW)

The following flowchart illustrates the strict sequential progression from raw user input to final intelligence dissemination across the frontend, backend, external STAC APIs, and export services:



3. SYSTEM COMPONENT ARCHITECTURE & INTER-PROCESS BRIDGES

FRONTEND LAYER (:8443)	REVERSE PROXY & GATEWAY	BACKEND & DATA LAYERS (:8000)
[React Frontend :8443] • React 18 + TypeScript • Tailwind CSS + Lucide • Leaflet & React-Leaflet • State: selectedArea • Pages: Search, Map, Compare, Analysis, Report	[Vite Dev Proxy] • Intercepts /api/* • Proxy Target: http://127.0.0.1:8000 • Eliminates CORS restrictions [FastAPI Gateway :8000] • GET /api/health • GET /api/dashboard • POST /api/search • POST /api/analysis	[Pydantic Validation (schemas.py)] • SearchRequest · SearchFilters • SearchResult · SelectedArea • AnalysisResponse · Finding [Backend Services (services.py)] • Location & AOI Resolver • Copernicus STAC Search Engine • Deterministic Analysis Engine • Master Catalog Fallback [Copernicus STAC API] • sentinel-2-l2a Collection

4. DETAILED SEARCH WORKFLOW & STAC DISCOVERY

The search subsystem serves as the semantic entry point for operators. Rather than requiring users to memorize satellite pass schedules or Military Grid Reference System (MGRS) tile identifiers, the system seamlessly maps human-readable place names and intent keywords to exact geospatial coordinates and STAC queries.

Step A: UI Query Dispatch	User submits query in SearchPage.tsx. The searchService.ts invokes POST /api/search with JSON payload { query : string, filters : SearchFilters }.
Step B: FastAPI Routing	The /api/search handler in main.py validates the payload against Pydantic schema SearchRequest and forwards execution to search_observations() in services.py.
Step C: AOI & Coordinate Resolution	resolve_location_and_aoi() evaluates text against curated catalogs (Pune, Hyderabad, Odisha, Gujarat, Chhattisgarh) and the 100+ entry INDIAN_GEO_LOCATIONS directory, producing bounding box [min_lon , min_lat , max_lon , max_lat].
Step D: Copernicus STAC Query	query_copernicus_stac() constructs an ISO datetime interval and cloud limit payload, issuing an HTTP/2 POST request to https://stac.dataspace.copernicus.eu/v1/search for collection sentinel-2-l2a .
Step E: Live STAC Response Parsing	Real ESA Sentinel-2 Level-2A granules are parsed. Metadata extracted includes product ID, acquisition timestamp, cloud cover percentage, sun elevation angle, relative orbit, and exact granule bounds.
Step F: Resilient Fallback Protection	If external STAC connectivity is unavailable or times out, the engine gracefully reverts to local curated catalogs (select_result_set). Non-existent locations cleanly return empty arrays [] (avoiding false fallbacks).
Step G: UI Projection & Marker Plotting	Frontend receives SearchResult[] , rendering dynamic observation cards, confidence badges, and Leaflet interactive circle markers centered on the satellite coordinates.

5. REAL VS. MOCKED FUNCTIONALITY MATRIX (AUDIT VERIFIED)

To maintain absolute academic and technical honesty before college faculty and hackathon evaluators, the table below strictly delineates between **live**, **production-verified functionality** and **simulated or mock components**:

SYSTEM SUBSYSTEM	STATUS	ACTUAL IMPLEMENTATION DETAILS & CODEBASE EVIDENCE
React Frontend Architecture	REAL	React 18 + TypeScript + Tailwind CSS with tab routing, Lucide icons, and responsive layout.
FastAPI Backend Server	REAL	FastAPI 0.141 running on Uvicorn (Port 8000) with strict Pydantic v2 schemas and CORS middleware.
Vite Network Proxy	REAL	Vite dev server (Port 8443) actively proxies /api/* requests to port 8000, eliminating CORS issues.
Copernicus STAC Connectivity	REAL	Live HTTP/2 integration with https://stac.dataspace.copernicus.eu/v1/search for sentinel-2-l2a .
Sentinel-2 Scene Metadata	REAL	Extracts live ESA scene IDs, acquisition dates, cloud cover, sun angles, orbits, and bounding boxes.
Location / AOI Resolution	REAL	Dynamic coordinate matching across 100+ Indian geographic regions and curated testbeds in services.py.
GeoJSON Export Engine	REAL	exportAreaToGeoJSON() generates RFC 7946 compliant FeatureCollections with bounding polygon geometry.
Print & Report Dossier	REAL	Browser print driver execution generates standardized, formatted mission intelligence dossiers.
Curated Fallback Datasets	MOCK / SIM	Static fallback observations (e.g. Pune, Hyderabad) ensure system reliability when STAC is unreachable.
Change Analysis Calculations	MOCK / SIM	calculate_analysis() uses deterministic mathematical ratios rather than pixel-level GeoTIFF raster math.
Spectral Imagery Visuals	MOCK / SIM	NDVI/false-color visual toggles apply client-side CSS filter transforms rather than raw band computations.
Before / After Map Imagery	MOCK / SIM	Split-slider visualizer currently renders web satellite basemaps rather than two distinct Sentinel-2 raster scenes.

6. CURRENT CHANGE ANALYSIS ENGINE VS. PLANNED RASTER PIPELINE

The repository currently separates high-level workflow validation from deep raster math. In the current state, the system demonstrates how satellite intelligence is packaged and delivered to decision-makers. The section below contrasts the operational deterministic calculation engine with the next-stage pixel pipeline.

CURRENT OPERATIONAL PIPELINE (Deterministic Calculation)	NEXT DEVELOPMENT STAGE (Planned Pixel-Level Raster Differencing)
1. Selected AOI Binding: User confirms area of interest. 2. calculate_analysis() Dispatch: Dispatched to FastAPI. 3. Deterministic Parsing: Area string parsed (e.g. '12.4 km²'). 4. Mathematical Vector Modeling: • Built-up before: <code>round(area_km * 0.65, 1)</code> • Built-up after: <code>round(built_before + area_km, 1)</code> • Net percentage: <code>+X.X%</code> 5. Intelligence Assembly: Generates structured findings list, telemetry statistics tuples, and chronologically ordered timelines. 6. Report & Dashboard Visualization: Rendered in UI.	1. Dual Scene Acquisition: Retrieve T_1 (baseline) & T_2 (comparison) Sentinel-2 L2A granules from CDSE. 2. Raster Band Extraction: Extract Band 4 (Red), Band 8 (NIR), and Band 3 (Green) at 10m spatial resolution. 3. Spectral Index Calculation: • Compute $NDVI(T_1)$ and $NDVI(T_2)$ • Compute $NDWI(T_1)$ and $NDWI(T_2)$ 4. Pixel-Level Differencing: $\Delta = Index(T_2) - Index(T_1)$. 5. Thresholding & Morphological Filtering: Segment noise. 6. Polygon Vectorization: Polygonize changed raster masks. 7. Geospatial Output: Exact change masks & GeoJSON.
Status: Fully Operational in Repository (Phase 1)	Status: Priority 2–6 Next Development Roadmap

7. MULTI-TEMPORAL CHANGE DETECTION METHODOLOGY (REMOTE SENSING)

Multi-temporal satellite change analysis is rooted in biophysical surface reflectance physics. When surface materials undergo transition (vegetation loss, urban construction, water level fluctuation), their spectral reflectance curves shift measurably across discrete satellite wavelength bands.

SPECTRAL INDEX	MATHEMATICAL FORMULA	SENTINEL-2 BANDS USED	PHYSICAL INTERPRETATION & THRESHOLDS
NDVI (Normalized Difference Vegetation Index)	$NDVI = (NIR - RED) / (NIR + RED)$ $NDVI\ Change = NDVI(T_2) - NDVI(T_1)$	Band 8 (NIR: 842 nm) Band 4 (Red: 665 nm) Resolution: 10 meters	Healthy chlorophyll strongly absorbs Red and reflects NIR. Negative change (NDVI Change < -0.15) signifies canopy thinning, clearcutting, deforestation, or crop harvesting stress.
NDWI (Normalized Difference Water Index)	$NDWI = (GREEN - NIR) / (GREEN + NIR)$ $NDWI\ Change = NDWI(T_2) - NDWI(T_1)$	Band 3 (Green: 560 nm) Band 8 (NIR: 842 nm) Resolution: 10 meters	Open water strongly absorbs NIR while reflecting visible Green. Positive values (> 0.0) classify water bodies. NDWI Change > +0.20 detects flood inundation; negative shifts indicate reservoir drying and shoreline recession.
MNDWI / NDBI (Modified Water / Built-Up Indices)	$MNDWI = (GREEN - SWIR) / (GREEN + SWIR)$ $NDBI = (SWIR - NIR) / (SWIR + NIR)$	Band 3 (Green: 560 nm) Band 11 (SWIR: 1610 nm) Band 8 (NIR: 842 nm)	SWIR wavelengths suppress urban building reflectance while enhancing open water delineation. NDBI positive values identify newly constructed asphalt, concrete, and built-up structures.

8. BEFORE / AFTER COMPARISON: CURRENT VS. INTENDED WORKFLOW

Current Visualizer State: The frontend provides an interactive split-screen comparison slider with synchronized Leaflet map viewports, panning, and zoom locks. In the current demo stage, the background basemap displays high-resolution public satellite tile layers with CSS-based spectral contrast adjustments.

Intended Production Workflow: In the upcoming phase, the comparison slider will directly bind to two distinct Copernicus Sentinel-2 L2A raster scenes: T_1 (Baseline Date) on the left pane and T_2 (Recent Date) on the right pane, overlaying automated change polygons generated from pixel-level differencing.

9. API SPECIFICATION & COMMUNICATION BRIDGE TABLE

ENDPOINT & METHOD	OPERATIONAL PURPOSE	FRONTEND CALLER	BACKEND HANDLER & SCHEMA
GET /api/health	Backend liveness and telemetry check; verifies server is online.	App.tsx on initial mount	<code>health_check()</code> in <code>main.py</code> Returns: <code>{ status, service, version }</code>
GET /api/dashboard	Retrieves national telemetry stats, monitored AOIs, and recent events.	App.tsx (Dashboard View)	<code>get_dashboard_data()</code> in <code>services.py</code> Returns: <code>DashboardResponse</code>
POST /api/search	Executes natural language / structured search with live STAC retrieval.	<code>searchService.ts</code> (<code>SearchPage.tsx</code>)	<code>search_observations(query, filters)</code> Input: <code>SearchRequest</code> Returns: <code>List[SearchResult]</code>
POST /api/analysis	Generates quantitative change metrics, severity, and findings for selected AOI.	<code>analysisService.ts</code> (<code>Analysis/Report Pages</code>)	<code>calculate_analysis(area)</code> in <code>services.py</code> Input: <code>SelectedArea</code> Returns: <code>AnalysisResponse</code>

10. OFFLINE HACKATHON DEPLOYMENT PLAN (AIR-GAPPED OPERATION)

CRITICAL EVALUATION ARCHITECTURE: ZERO FULL-ARCHIVE DOWNLOAD A common misconception in satellite hackathons is attempting to mirror the entire multi-terabyte Copernicus archive locally. For the SIH live presentation and internal hackathon defense, our architecture implements an air-gapped caching strategy : only 3 to 5 targeted demonstration AOIs (e.g. Pune Urban Fringe, Hyderabad Outer Ring, Hasdeo Arand Forest) have their exact Sentinel-2 Level-2A GeoTIFF tiles and metadata pre-cached locally. The entire application runs 100% offline without public internet.	
1. Pre-Cache Target AOIs	Download 3–5 representative pairs of Sentinel-2 L2A granules for selected demonstration testbeds.
2. Local Directory Structure	Store GeoTIFF bands (B02, B03, B04, B08) in <code>SIH_BACKEND/data/cached_scenes/{scene_id}/</code> .
3. Local STAC Simulation	FastAPI intercepts STAC queries and serves pre-cached metadata instantly with zero network latency.
4. Embedded Raster Engine	Python <code>rasterio</code> and <code>numpy</code> process local raster arrays in under 2 seconds on standard laptop hardware.
5. Offline Basemap Tiles	Frontend utilizes local MBTiles or pre-rendered Leaflet tile bundles, eliminating external map dependencies.

11. CURRENT LIMITATIONS & GAP ANALYSIS

Limitation Area	Current Repository Behavior	Technical Gap & Resolution Path
Semantic Search Engine	Keyword matching + Regex + STAC filter	Does not yet execute deep visual-semantic embeddings (CLIP/RemoteCLIP + FAISS). Planned for Priority 8.
Change Analysis Engine	Deterministic mathematical modeling	Does not perform pixel-level GeoTIFF raster subtraction. Next stage will integrate rasterio array differencing.
Observation Provenance	Live STAC metadata + Curated Fallbacks	Curated catalog entries provide safety when offline, but represent simulated baseline records.
Temporal Visualizer	Synchronized dual basemap tiles	Comparison view uses web map tiles rather than two distinct Sentinel-2 raster products.
Geocoding Granularity	100+ Major Indian Cities & States	Arbitrary remote rural villages require integration with Nominatim / Bhuvan geocoding services.

12. NEXT DEVELOPMENT PRIORITIES (ROADMAP FOR HACKATHON)

Priority & Milestone	Objective & Implementation Target	Target Deadline
Priority 1: Geographic Resolution	Enhance multi-tier location matching to resolve arbitrary Indian sub-districts and talukas.	Sep 14, 2026
Priority 2: Temporal Pair Retrieval	Automate retrieval of paired Sentinel-2 scenes for exact Before (T_1) and After (T_2) timestamps.	Sep 15, 2026
Priority 3: Raster Band Extraction	Download/read required raw bands (Band 2, 3, 4, 8) as NumPy arrays via rasterio .	Sep 16, 2026
Priority 4: Pixel-Level NDVI/NDWI Math	Execute matrix-level differencing: $\text{Index Change} = \text{Index}(T_2) - \text{Index}(T_1)$ to isolate changed pixels.	Sep 17, 2026
Priority 5: Real Imagery Rendering	Feed true T_1 and T_2 RGB/false-color GeoTIFF renders into the frontend split comparison slider.	Sep 18, 2026
Priority 6: Actual Vector Polygons	Vectorize connected changed pixel masks into GeoJSON polygons with exact area square kilometers.	Sep 19, 2026
Priority 7: Offline Demo Caching	Bundle 5 key Indian testbeds locally for guaranteed 100% offline hackathon demonstration.	Sep 19, 2026
Priority 8: Visual Embeddings (CLIP)	Long-term integration of RemoteCLIP + FAISS for zero-shot text-to-satellite semantic retrieval.	Post-Hackathon

13. STEP-BY-STEP DEMONSTRATION FLOW FOR EVALUATORS

During live project defense before faculty members and hackathon judges, follow this sequential 9-step demonstration:

1. Query Submission:	Navigate to Search tab. Type 'Urban development in Hyderabad' or select Maharashtra filter.
2. Live STAC Verification:	Point out live Sentinel-2 metadata returned (real granule ID, cloud cover %, orbit, sun angle).
3. AOI Selection:	Click on a returned satellite observation tile. Demonstrate Leaflet map fly-to animation and boundary plot.
4. Temporal Baseline Binding:	Note the Before/After observation timestamps synchronized into application state.
5. Satellite Imagery Inspection:	Switch to Map Page. Demonstrate layer toggles (NDVI, NDWI, False Color, True Color).
6. Run Change Detection:	Click 'Run Analysis'. Demonstrate sub-second dispatch to FastAPI /api/analysis endpoint.
7. Metric Evaluation:	Review quantitative impact cards (Net Change, % Expansion, Severity Rating, Historical Timeline).
8. Dual Viewport Comparison:	Open Comparison view. Drag split slider to visually compare baseline vs. post-event imagery.
9. Intelligence Dissemination:	Click 'Export GeoJSON' (downloads valid RFC 7946 polygon) and 'Export Dossier' (printable report).

14. FINAL SUMMARY: END-TO-END PIPELINE & ARCHITECTURAL EVOLUTION

This concluding summary synthesizes the entire project lifecycle into a unified technical diagram, followed by a side-by-side architectural evolution matrix contrasting our current implementation with the target deployment.

USER	Natural Language Query / Temporal & Geographic Filters
FRONTEND	React 18 · TypeScript · Vite (:8443) · Leaflet Map Engine
FASTAPI	Uvicorn Gateway (:8000) · Pydantic Request/Response Validation
LOCATION / AOI	services.py · Indian Geographic Catalog Resolver & Bounding Box
COPERNICUS STAC	https://stac.dataspace.copernicus.eu/v1/search (HTTP/2 Client)
SENTINEL-2 L2A	Granule Discovery · Real Acquisition Dates, Cloud Mask & Sun Angles
BEFORE + AFTER	Dual Temporal Observation Binding (T_1 Baseline vs T_2 Comparison)
RASTER ANALYSIS	Current: Deterministic Modeling Target: NumPy/Rasterio Differencing
CHANGE DETECTION	Spectral Index Extraction (NDVI / NDWI) & Vector Polygon Masking
VISUALIZATION	Interactive Split Comparison Slider & Spectral Layer Filters
INTELLIGENCE REPORT	RFC 7946 GeoJSON Export & Printable Mission Dossier

Current Status vs. Target Architecture Matrix:

ARCHITECTURAL LAYER	CURRENT IMPLEMENTATION (SEPT 2026)	TARGET PRODUCTION ARCHITECTURE
Satellite Data Source	Live Copernicus STAC metadata + Curated Fallback Catalogs	Direct automated Sentinel-2 / Cartosat GeoTIFF band streaming
Search Methodology	Keyword intent matching + Geo Directory coordinate lookup	Multi-modal RemoteCLIP visual embeddings + FAISS vector index
Change Computation	Deterministic mathematical modeling of change vectors	Pixel-level NumPy raster math ($T_2 - T_1$) with morphological filters
Visual Comparison	Synchronized dual viewport with CSS spectral filter styling	True calibrated T1/T2 GeoTIFF imagery rendering in Leaflet
Geospatial Vector Output	Client-side GeoJSON bounding polygon calculation	Automated polygonization of detected changed pixel boundaries
Deployment Readiness	Verified local full-stack (Zero compile/type errors)	Air-gapped offline demo package with pre-cached Indian testbeds

ACADEMIC INTEGRITY & HACKATHON READINESS STATEMENT

This document accurately reflects the operational reality of the SIH Problem Statement 26227 repository. The project has successfully completed **Phase 1**: establishing a resilient, error-free full-stack architecture connected to live European Space Agency (ESA) Sentinel-2 Level-2A data via the Copernicus Data Space STAC API. The path toward Phase 2 (pixel-level raster change detection) is strictly mapped out for implementation prior to the national finals.